

Total Score

270/320

Points Achieved

Overall Percentage

84.4%

Excellent Performance

Final Grade

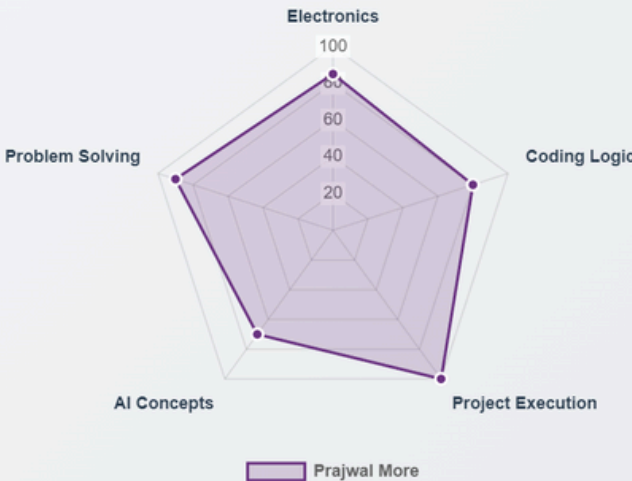
A

Grade 6 Standard

Completion

100%

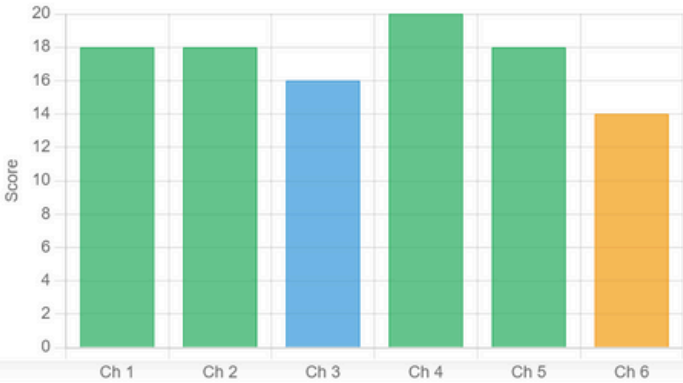
All Modules Completed



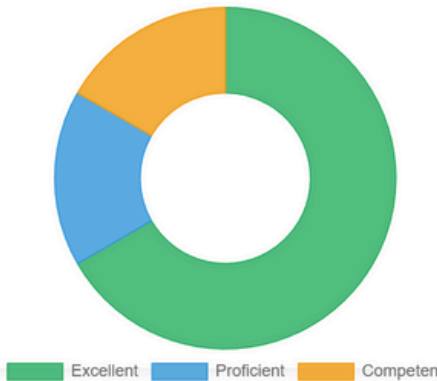
Arduino Nano From Basic To Brilliance

104/120 (86.7%)

Chapter Performance



Skill Distribution



Intro to Arduino Nano

18/20

Excellent - Solid understanding of board fundamentals

LEDs & Simple Circuits

18/20

Excellent - Good circuit safety practices

Sensors & Inputs

16/20

Proficient - Needs practice with potentiometer logic

Sound & Temperature

20/20

Outstanding - Perfect execution of input/output logic

Servos & Displays

18/20

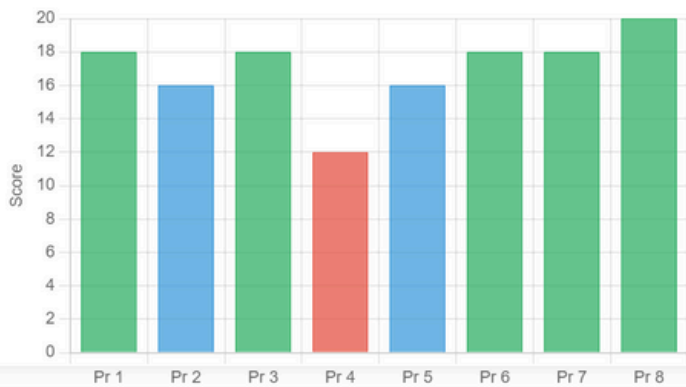
Excellent - Strong motor control understanding

Ultrasonic Sensors

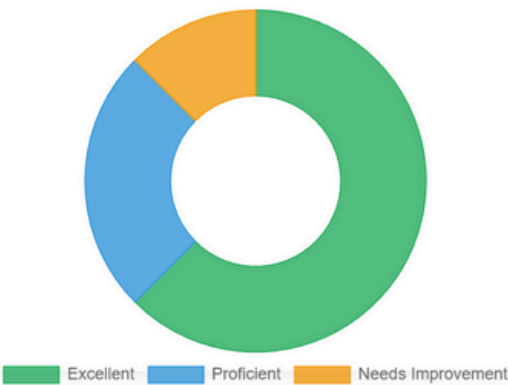
14/20

Competent - Confused pulseIn() with analogRead()

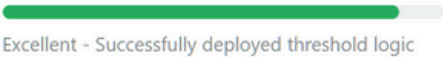
Project Performance



Skill Areas



LDR Security Alarm 18/20



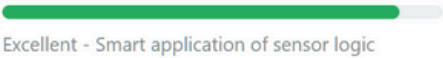
Excellent - Successfully deployed threshold logic

Button Piano 16/20



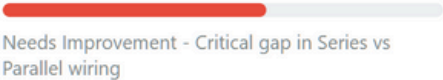
Proficient - Good array usage, struggled with LED integration

Hand Wave Light Switch 18/20



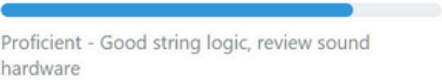
Excellent - Smart application of sensor logic

LDR Bar Graph 12/20



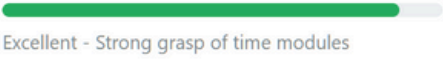
Needs Improvement - Critical gap in Series vs Parallel wiring

Morse Code Decoder 16/20



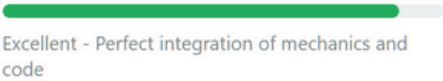
Proficient - Good string logic, review sound hardware

Digital Clock (RTC) 18/20



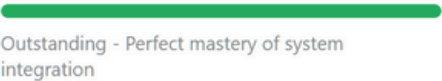
Excellent - Strong grasp of time modules

Auto Gate Opener 18/20



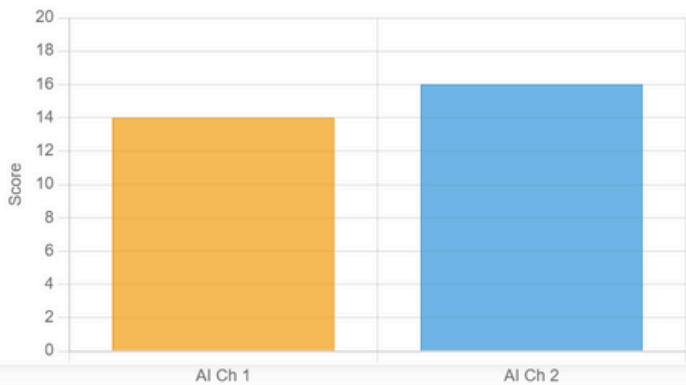
Excellent - Perfect integration of mechanics and code

Motion Detector Alarm 20/20



Outstanding - Perfect mastery of system integration

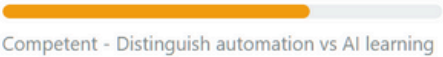
AI Chapter Performance



Understanding Levels

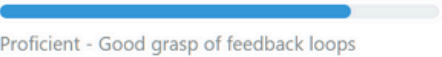


How AI Solves Problems 14/20



Competent - Distinguish automation vs AI learning

How AI Learns 16/20



Proficient - Good grasp of feedback loops

STUDENT PROFILE

Name: Prajwal More

Grade: 6th

Roll No: 21

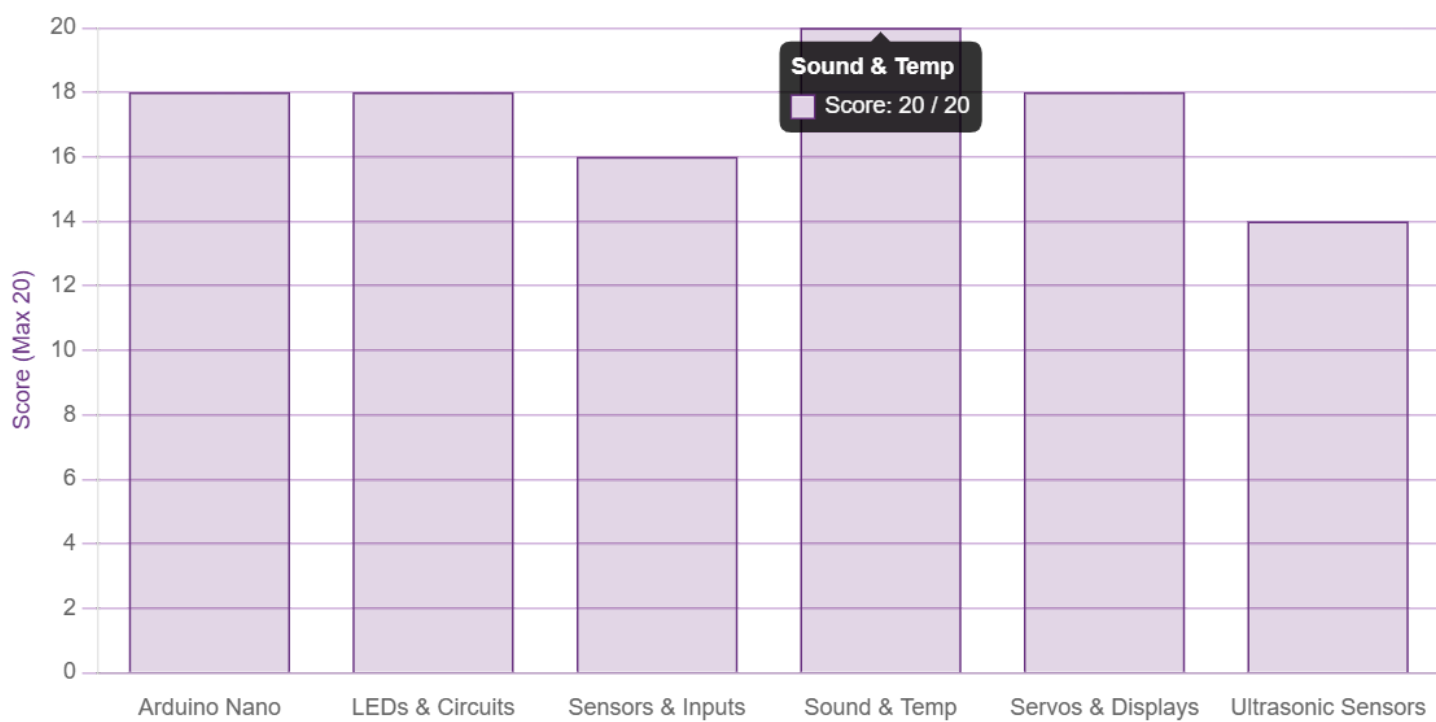
Division: A

Course: Arduino Nano From Basic To Brilliance

MODULE TOTAL: 104 / 120 (86.7%) — Grade: A (Excellent)

CHAPTER BREAKDOWN: Arduino Nano From Basic To Brilliance

Chapter Scores out of 20



****Focus Area:**** Distinguishing between Analog and Digital commands/readings (e.g., `pulseIn()` vs. `analogRead()`).

Chapter/Module	Score	Status	Feedback Summary
Intro to Arduino Nano	18/20	Excellent	Solid understanding of the board. Review power pin identification.
LEDs & Simple Circuits	18/20	Excellent	Great safety habits. **Note:** Remember standard resistor is 220Ω.
Sensors & Inputs	16/20	Proficient	Needs practice on Potentiometer logic.
Sound & Temperature	20/20	Outstanding	Flawless execution of input/output logic.

Chapter/Module	Score	Status	Feedback Summary
Servos & Displays	18/20	Excellent	Strong motor control logic.
Ultrasonic Sensors	14/20	Competent	Confused `pulseIn()` (time) with `analogRead()` (voltage).

Program Director
WE Smart Lab

Date of Report
November 22, 2024

 STUDENT PROFILE

Name: Prajwal More

Grade: 6th

Division: A

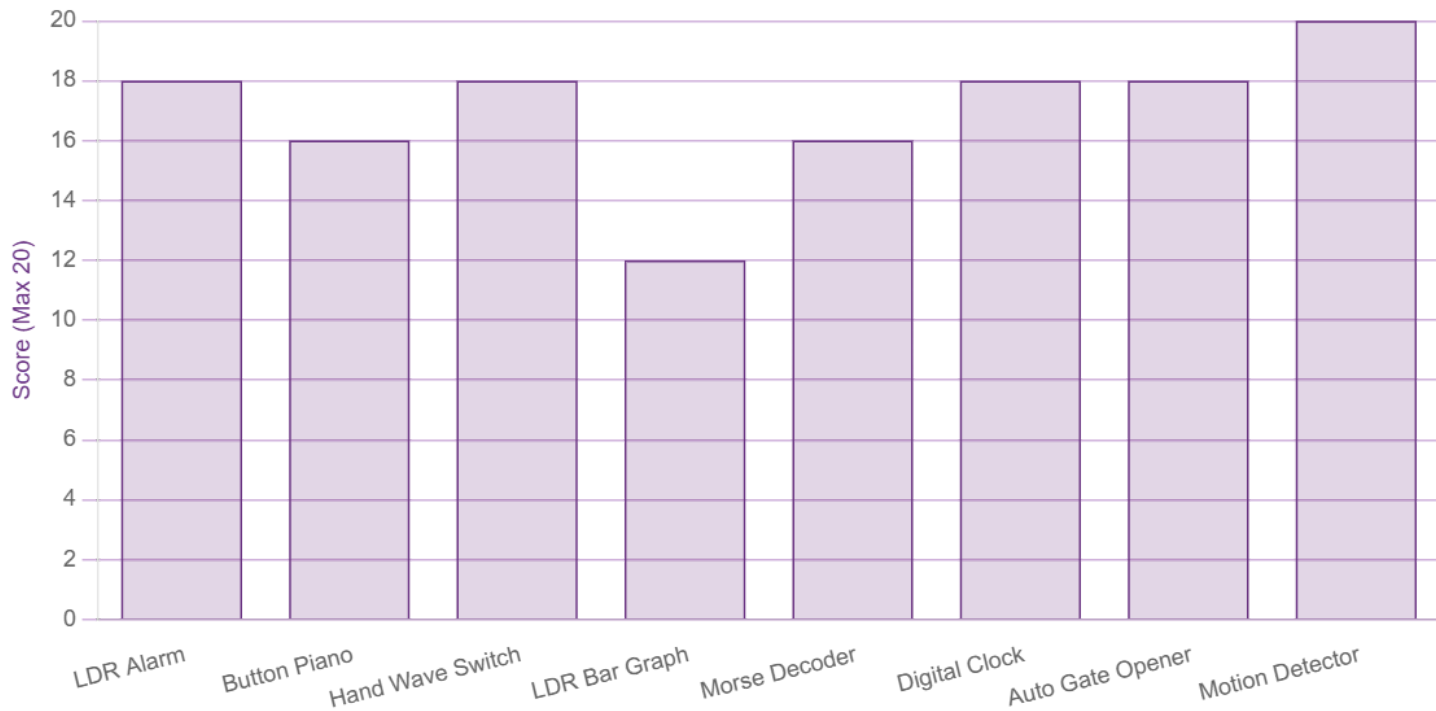
Roll No: 21

Course: Arduino Nano From Basic To Brilliance

MODULE TOTAL: 136 / 160 (85.0%) — Grade: A (Excellent)

PROJECT BREAKDOWN: PRACTICAL APPLICATION

Project Scores out of 20



****Critical Gap:**** Confusion between ****Series vs. Parallel wiring**** (e.g., LDR Bar Graph project).

Project Module	Score	Status	Feedback Summary
LDR Security Alarm	18/20	Excellent	Successfully deployed threshold logic.
Button Piano	16/20	Proficient	Struggled with adding independent LEDs to the circuit.
Hand Wave Light Switch	18/20	Excellent	Smart application of sensor logic.
LDR Bar Graph	12/20	Needs Improvement 	**Critical Gap:** Confused Series vs. Parallel wiring for multi-LED displays.

Project Module	Score	Status	Feedback Summary
Morse Code Decoder	16/20	Proficient	Good string logic. Hardware integration for sound needs review.
Digital Clock (RTC)	18/20	Excellent	Excellent grasp of time modules.
Auto Gate Opener	18/20	Excellent	Perfect integration of mechanics (Servo) and code.
Motion Detector Alarm	20/20	Outstanding 🌟	Perfect score. Demonstrated complete mastery of system integration.

 STUDENT PROFILE

Name: Prajwal More

Grade: 6th

Division: A

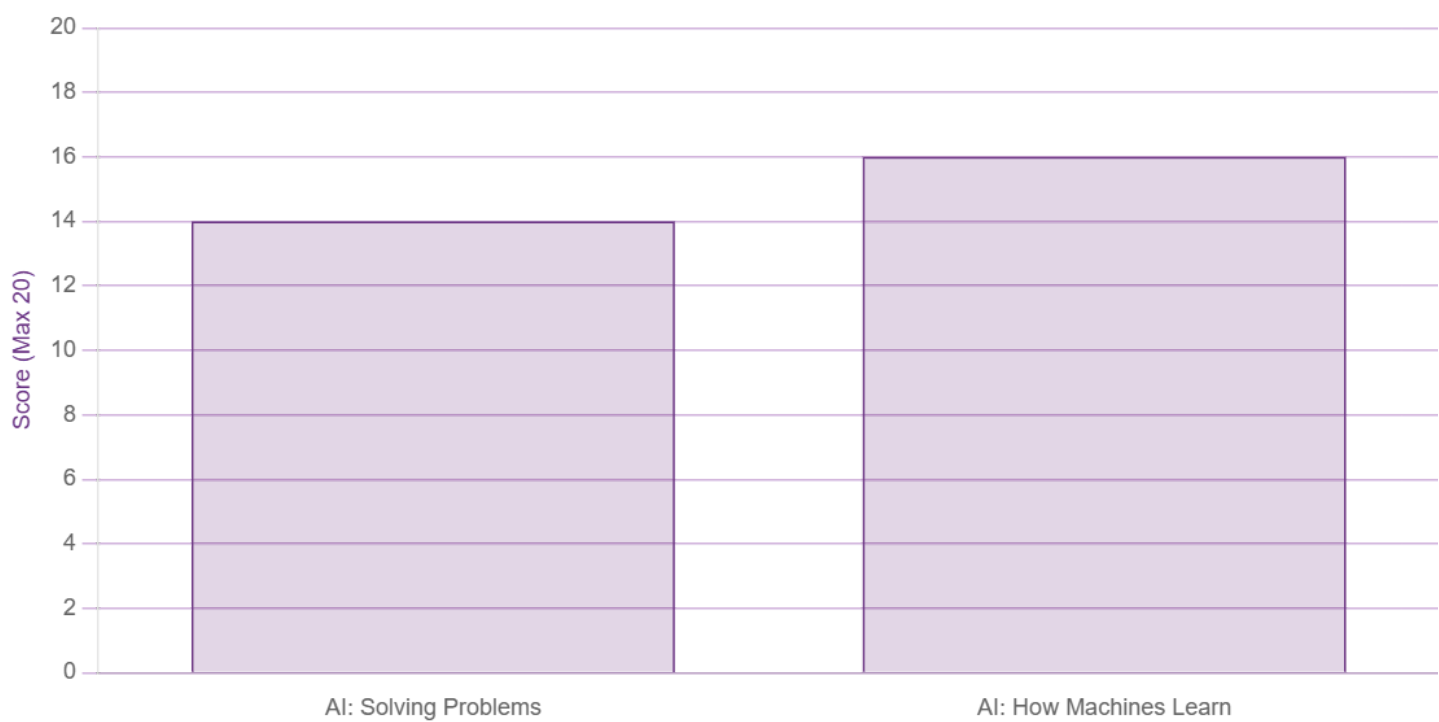
Roll No: 21

Course: Arduino Nano From Basic To Brilliance

MODULE TOTAL: 30 / 40 (75.0%) — Grade: B (Proficient)

CHAPTER BREAKDOWN: AI CONCEPTS (THEORY)

Chapter Scores out of 20



****Focus Area:**** Defining "Smart" (Automation) vs. "AI" (Learning/Data-driven).

Chapter/Module	Score	Status	Feedback Summary
AI: Solving Problems	14/20	Competent	Needs to better distinguish between basic automation and AI learning.
AI: How Machines Learn	16/20	Proficient	Good grasp of the "feedback loop" concept.

☐ Student Details :

School Name: Veer Baji Prabhv Vidyalay

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature : [Signature]

☐ Marks Distribution: ☒ Question : 10 ☐ Total : 20 Marks

Chapter 1: Introduction to Arduino Nano

18
20

1. What is the primary function of the Arduino Nano?

- ☐ a) To act as a power source
- ☒ b) To control electronic components based on programmed instructions
- ☒ c) To display images on a screen
- ☐ d) To connect to the internet

2. Which pin on the Arduino Nano is typically used for PWM (Pulse Width Modulation)?

- ☐ a) A0
- ☐ b) D13
- ☒ c) D9 (marked with ~)
- ☐ d) GND

3. What does the "pinMode(2, OUTPUT)" command do?

- ☐ a) Sets pin 2 to read input signals
- ☒ b) Sets pin 2 to send output signals
- ☐ c) Turns pin 2 off
- ☒ d) Resets pin 2

4. Which component is essential for protecting an LED in a circuit with the Arduino Nano?

- ☐ a) Capacitor
- ☒ b) Resistor
- ☐ c) Transistor
- ☐ d) Diode

5. What does the "void loop()" function do in an Arduino sketch?

- ☐ a) Runs once when the Arduino starts
- ☐ b) Turns off the Arduino
- ☒ c) Repeats continuously while the Arduino is powered
- ☐ d) Connects to the internet

6. Which Arduino Nano pin is connected to the built-in LED?

- ☐ a) D1
- ☒ b) D13
- ☐ c) A0
- ☐ d) 5V

7. What is the purpose of the "delay(1000)" command?

- ☐ a) Pauses the program for 1 second
- ☒ b) Speeds up the program
- ☐ c) Turns off all pins
- ☐ d) Measures voltage

8. Which of the following is NOT a power pin on the Arduino Nano?

- ☐ a) 3V3
- ☐ b) 5V
- ☐ c) GND
- ☒ d) D7

9. What does the "digitalWrite(2, HIGH)" command do?

- ☐ a) Turns pin 2 off
- ☐ b) Sets pin 2 to input mode
- ☒ c) Sends a high voltage (5V) to pin 2
- ☐ d) Measures voltage on pin 2

10. Which tool is used to write and upload code to the Arduino Nano?

- ☒ a) Arduino IDE
- ☐ b) Microsoft Word
- ☐ c) Photoshop
- ☐ d) Web browse

☐ Student Details :

School Name: Veer Baji Prabhur Vidyalay

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature : *[Signature]*

☐ Marks Distribution: ☒ Question : 10 ☐ Total : 20 Marks

Chapter 2: Exploring LEDs and Simple Circuits

1. What does LED stand for?

- ☒ a) Light Emitting Diode
- ☐ b) Low Energy Device
- ☐ c) Long Electric Display
- ☐ d) Light Energy Detector

2. Why is a resistor used in series with an LED?

- ☐ a) To increase brightness
- ☒ b) To limit current and prevent damage
- ☒ c) To change the LED color
- ☐ d) To reduce voltage

3. Which Arduino function adjusts LED brightness using PWM?

- ☐ a) digitalRead()
- ☒ b) analogWrite()
- ☒ c) pinMode()
- ☐ d) delay()

4. In a traffic light simulator, which LED typically represents "stop"?

- ☐ a) Green
- ☐ b) Yellow
- ☒ c) Red
- ☐ d) Blue

5. What is the purpose of the breadboard in a circuit?

- ☐ a) To write code
- ☐ b) To power the Arduino
- ☒ c) To connect components without soldering
- ☐ d) To display output

18
20

6. Which command turns an LED off?

- ☐ a) digitalWrite(pin, HIGH)
- ☒ b) digitalWrite(pin, LOW)
- ☐ c) analogWrite(pin, 255)
- ☐ d) pinMode(pin, INPUT)

7. What does PWM stand for?

- ☒ a) Pulse Width Modulation
- ☐ b) Power Wave Management
- ☐ c) Programmable Wiring Mode
- ☐ d) Pin Wiring Method

8. Which component is NOT needed for a basic LED circuit?

- ☐ a) Resistor
- ☐ b) Jumper wires
- ☒ c) Servo motor
- ☐ d) LED

9. How many LEDs are used in a traffic light simulator?

- ☒ a) 1
- ☐ b) 2
- ☒ c) 3
- ☐ d) 4

10. What is the typical resistance value for an LED circuit?

- ☐ a) 220 ohms
 - ☐ b) 10k ohms
 - ☒ c) 1 ohm
 - ☐ d) 100k ohms
-

☐ Student Details :

School Name: Veer Baiji Prabhu Vidyalay

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature : Roddy

☐ Marks Distribution: ☒ Question : 10  Total :20 Marks

Chapter 3: Sensors and Inputs

16
20

1. What is the difference between digital and analog inputs?

- ☒ a) Digital reads only HIGH/LOW; analog reads a range
- ☐ b) Digital reads a range; analog reads only HIGH/LOW
- ☐ c) Both are the same
- ☐ d) Digital is faster

2. Which sensor measures light intensity?

- ☐ a) Ultrasonic sensor
- ☒ b) LDR (Light Dependent Resistor)
- ☐ c) Temperature sensor
- ☐ d) Servo motor

3. What does an LDR-based night lamp do?

- ☐ a) Turns on in bright light
- ☒ b) Turns on in darkness
- ☐ c) Measures temperature
- ☐ d) Controls a servo

4. Which Arduino function reads analog inputs?

- ☒ a) digitalWrite()
- ☐ b) delay()
- ☐ c) pinMode()
- ☐ d) analogRead()

5. What is the output range of analogRead()?

- ☐ a) 0-1023
- ☒ b) 0-255
- ☐ c) 0-5
- ☐ d) 0-100

6. Which component is used to create variable resistance in a circuit?

- ☐ a) LED
- ☒ b) Potentiometer
- ☐ c) Buzzer
- ☐ d) Servo

7. What does a push button connected to an Arduino Nano act as?

- ☐ a) Analog input
- ☐ b) PWM output
- ☒ c) Digital input
- ☐ d) Power source

8. How does a potentiometer control an LED's brightness?

- ☐ a) By sending PWM signals
- ☒ b) By changing resistance and analog input
- ☐ c) By turning the LED off
- ☐ d) By measuring temperature

9. What is the purpose of the "map()" function?

- ☒ a) To convert one range of values to another
- ☐ b) To turn pins on/off
- ☐ c) To measure distance
- ☐ d) To control servos

10. Which sensor detects distance using sound waves?

- ☒ a) Ultrasonic sensor (HC-SR04)
- ☐ b) LDR
- ☐ c) Temperature sensor
- ☐ d) Potentiometer

☐ Student Details :

School Name: Neer Baji Prabhu Vidyaly

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature: *[Signature]*

☐ Marks Distribution: ☒ Question : 10  Total :20 Marks

Chapter 4: Sound and Temperature

20
20

1. What does a piezo buzzer produce?

- ☐ a) Light
- ☒ b) Sound
- ☐ c) Heat
- ☐ d) Motion

2. Which Arduino function generates a tone?

- ☐ a) digitalWrite()
- ☒ b) tone()
- ☐ c) analogRead()
- ☐ d) delay()

3. What does a temperature sensor measure?

- ☐ a) Light intensity
- ☐ b) Sound waves
- ☒ c) Heat
- ☐ d) Distance

4. How is temperature data displayed in Project 4.3?

- ☒ a) On an LCD screen
- ☐ b) Using Morse code
- ☐ c) Via a servo motor
- ☐ d) Through an LED

5. What triggers the buzzer in the temperature alarm project?

- ☐ a) Low light
- ☒ b) High temperature
- ☐ c) Button press
- ☐ d) Servo movement

6. Which library is used for the 4-digit display?

- ☐ a) Servo.h
- ☒ b) TM1637.h
- ☐ c) LiquidCrystal.h
- ☐ d) Wire.h

7. What is the purpose of the "Serial.begin(9600)" command?

- ☒ a) To start serial communication for debugging
- ☐ b) To turn on an LED
- ☐ c) To control a servo
- ☐ d) To measure distance

8. Which component is NOT used in the temperature alarm project?

- ☐ a) Buzzer
- ☐ b) Temperature sensor
- ☒ c) Ultrasonic sensor
- ☐ d) LED

9. What does the "delayMicroseconds()" function do?

- ☒ a) Pauses the program in microseconds
- ☐ b) Measures temperature
- ☐ c) Controls servo angle
- ☐ d) Reads analog inputs

10. How is the temperature value converted from sensor data?

- ☐ a) Using map()
- ☒ b) Using mathematical formulas
- ☐ c) Using digitalWrite()
- ☐ d) It doesn't need conversion

☐ Student Details :

School Name: Veer Baiji Prabhu Vidyalaya
Name : Prajwal More
Standard : _____ Division : _____ Roll No : 21

Examiner's Signature : [Signature]

☐ Marks Distribution: ☒ Question : 10  Total :20 Marks

Chapter 5: Marvellous Movements with Servos and Displays

1. What is the primary function of a servo motor?

- ☐ a) To emit light
- ☒ b) To rotate to specific angles based on input
- ☐ c) To measure temperature
- ☐ d) To act as a power source

2. Which Arduino library is used to control a servo?

- ☒ a) Servo.h
- ☐ b) Wire.h
- ☐ c) LiquidCrystal.h
- ☐ d) TM1637.h

3. How is a servo motor's angle typically controlled?

- ☐ a) Using digitalWrite()
- ☐ b) Using servo.write(angle)
- ☒ c) Using analogRead()
- ☐ d) Using delay()

4. What does the map() function do in a servo project?

- ☐ a) Measures distance
- ☐ b) Turns the servo off
- ☒ c) Converts sensor values to angle ranges (e.g., 0–180°)
- ☐ d) Displays text on an LCD

5. Which component is used to manually adjust servo angle in Project 5.2?

- ☐ a) LDR
- ☐ b) Buzzer
- ☐ c) Ultrasonic sensor
- ☒ d) Potentiometer

18
20

6. What is the purpose of the 4-digit display in Project 5.3?

- ☐ a) To show temperature
- ☒ b) To display the servo's current angle
- ☐ c) To emit sound
- ☐ d) To measure light

7. Which pin on the Arduino Nano is commonly used for servo control?

- ☐ a) D9 (PWM-capable)
- ☐ b) A0
- ☒ c) D13
- ☐ d) 5V

8. What happens if you set `servo.write(90)`?

- ☒ a) The servo moves to 90 degrees (neutral position)
- ☐ b) The servo spins continuously
- ☐ c) The servo turns off
- ☐ d) The servo emits a tone

9. Which wire on a servo connects to ground?

- ☐ a) Red
- ☒ b) Brown/Black
- ☐ c) Yellow/Orange
- ☐ d) Blue

10. What is the typical angle range of a standard servo motor?

- ☐ a) 0–90°
- ☐ b) 0–360°
- ☒ c) 0–180°
- ☐ d) 0–255°

☐ Student Details :

School Name: Veer Baji Prabhoo Vidyalay

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature : *[Signature]*

☐ Marks Distribution: ☒ Question : 10  Total : 20 Marks

Chapter 6: Sensing the World Around You

14
20

1. What does an ultrasonic sensor measure?

- ☐ a) Light intensity
- ☒ b) Distance using sound waves
- ☐ c) Temperature
- ☐ d) Humidity

2. Which pins on the HC-SR04 ultrasonic sensor connect to the Arduino?

- ☒ a) VCC, GND, Trig, Echo
- ☐ b) VCC, GND, SDA, SCL
- ☐ c) A0, A1, D2, D3
- ☐ d) 5V, GND, PWM, RX

3. What is the purpose of the pulseIn() function in an ultrasonic project?

- ☐ a) Measures the duration of a pulse to calculate distance
- ☐ b) Controls servo angle
- ☒ c) Reads analog inputs
- ☐ d) Generates sound

4. How is distance calculated from ultrasonic sensor data?

- ☒ a) distance = duration / 1000
- ☐ b) distance = duration × 5
- ☐ c) distance = duration × 0.034 / 2
- ☐ d) distance = duration + 10

5. What component displays distance measurements in Project 6.3?

- ☐ a) LED
- ☒ b) LCD screen
- ☐ c) Buzzer
- ☐ d) 7-segment display

6. What triggers the buzzer in the ultrasonic alarm project?

- ☒ a) Objects closer than a set distance (e.g., 10 cm)
- ☐ b) High temperature
- ☐ c) Bright light
- ☐ d) Servo movement

7. Which library is used for the 7-segment display?

- ☐ a) Servo.h
- ☒ b) TM1637.h
- ☐ c) LiquidCrystal.h
- ☐ d) Wire.h

8. What does the NewPing library simplify?

- ☐ a) Temperature measurement
- ☐ b) Servo control
- ☒ c) Ultrasonic sensor coding
- ☐ d) LED brightness

9. In Project 6.1, what does the LED indicate?

- ☐ a) Temperature
- ☒ b) Proximity of an object (e.g., blinks if too close)
- ☐ c) Light intensity
- ☐ d) Servo angle

10. Which sensor is used in the motion detector alarm system?

- ☐ a) PIR sensor
- ☐ b) LDR
- ☒ c) Potentiometer
- ☐ d) Temperature sensor

☐ Student Details :

School Name: Veer Bajji Prabhu Vidyalay

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature : *Radha*

☐ Marks Distribution: ☒ Question : 10  Total :20 Marks

Project 1: LDR Security Alarm

1. What triggers the alarm in this project?

- ☐ a) High temperature
- ☒ b) Sudden drop in light levels (LDR reading < threshold)
- ☐ c) Loud sound
- ☐ d) Motion detection

2. Which component detects light changes?

- ☐ a) Ultrasonic sensor
- ☒ b) LDR (Light Dependent Resistor)
- ☐ c) Potentiometer
- ☐ d) Servo motor

3. How is the buzzer activated?

- ☒ a) digitalWrite(buzzerPin, HIGH) when light level is low
- ☐ b) analogWrite(buzzerPin, 255)
- ☐ c) Only when a button is pressed
- ☐ d) Continuously regardless of light

4. What is the purpose of the delay(500) in the loop?

- ☐ a) To control servo angle
- ☐ b) To measure temperature
- ☒ c) To prevent rapid toggling of the buzzer
- ☐ d) To reset the Arduino

5. Which pin mode is used for the LDR module's OUT pin?

- ☒ a) INPUT
- ☐ b) OUTPUT
- ☐ c) INPUT_PULLUP
- ☐ d) PWM

18
20

6. How would you adjust the project to make it less sensitive?

- ☐ a) Use digitalRead() instead of analogRead()
- ☐ b) Remove the resistor
- ☐ c) Replace the LDR with an LED
- ☒ d) Increase the threshold value (e.g., if (lightLevel < 200))

7. What happens if the LDR is covered with your hand?

- ☒ a) Buzzer turns on
- ☐ b) Servo moves to 90°
- ☐ c) LCD displays "Hello"
- ☐ d) Nothing happens

8. Which library is not needed for this project?

- ☐ a) LiquidCrystal.h
- ☒ b) None (uses only basic Arduino functions)
- ☐ c) Servo.h
- ☐ d) Wire.h

9. What type of signal does the LDR send to the Arduino?

- ☐ a) Analog voltage (0–1023)
- ☒ b) Digital HIGH/LOW
- ☐ c) PWM signal
- ☐ d) Serial data

10. How could you modify the project to add an LED indicator?

- ☐ a) Connect an LED to a digital pin and turn it on with the buzzer
 - ☐ b) Replace the buzzer with an LED
 - ☒ c) Use the built-in LED on pin 13
 - ☒ d) Both a and c
-

☐ Student Details :

School Name: Veer Baji Prabhhu Vidyaly

Name : Prajwal More

Standard : 6th Division : _____ Roll No : 21

Examiner's Signature : Radhu

☐ Marks Distribution: ☒ Question : 10  Total :20 Marks

Project 2: Button Piano

16
20

1. How are musical notes generated in this project?

- ☒ a) Using tone() with predefined frequencies
- ☐ b) By spinning a servo
- ☐ c) Using analogWrite()
- ☐ d) Via Morse code

2. What is the purpose of the while(digitalRead(buttonPin) == LOW) loop?

- ☐ a) Measure light levels
- ☐ b) Increase the note's volume
- ☒ c) Wait until the button is released
- ☐ d) Reset the Arduino

3. Which button component is not needed?

- ☐ a) Resistor (uses INPUT_PULLUP)
- ☒ b) Capacitor
- ☐ c) Jumper wires
- ☐ d) Breadboard

4. How many buttons are used for the 8-note piano?

- ☒ a) 8 (D3-D10)
- ☐ b) 1
- ☐ c) 3
- ☐ d) 12

5. What does noteFrequencies[] = {262, 294, 330...} represent?

- ☒ a) Frequencies for notes C4, D4, E4...
- ☐ b) Delay times in milliseconds
- ☐ c) PWM values for LEDs
- ☐ d) Temperature readings

6. Which function stops the buzzer sound?

- ☐ a) noTone()
- ☐ b) digitalWrite(buzzerPin, LOW)
- ☒ c) delay(1000)
- ☐ d) Both a and b

7. How would you add an LED that blinks with each note?

- ☐ a) Replace the buzzer with an LED
- ☐ b) Connect an LED to pin 13 and toggle it in the button loop
- ☒ c) Use analogWrite() on the buzzer pin
- ☐ d) It's impossible

8. What is the role of delay(100) in the loop?

- ☐ a) Debounce the button press
- ☐ b) Change the note's pitch
- ☒ c) Measure distance
- ☐ d) Power off the buzzer

9. Which musical note corresponds to 262 Hz?

- ☐ a) A4
- ☒ b) C4
- ☐ c) G5
- ☐ d) B3

10. How could you expand this project to play chords?

- ☒ a) Use multiple buzzers and press multiple buttons simultaneously
 - ☐ b) Replace the Arduino with a synthesizer
 - ☒ c) It's not possible
 - ☐ d) Use a servo to press buttons
-

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Project 3: Hand Wave Light Switch

1. Which sensor detects hand waves?

- ☐ a) LDR
- ☒ b) Ultrasonic (HC-SR04)
- ☐ c) Temperature sensor
- ☐ d) PIR sensor

2. What condition turns the LED on?

- ☒ a) if (distance < 20 && distance > 0)
- ☐ b) if (lightLevel > 500)
- ☒ c) if (temperature > 30)
- ☐ d) if (buttonPressed)

3. How is distance calculated from the ultrasonic sensor?

- ☒ a) duration × 0.034 / 2
- ☐ b) pulseIn() / 1000
- ☐ c) analogRead(A0)
- ☐ d) digitalRead(D2)

4. What does digitalWrite(TRIG_PIN, LOW) do?

- ☐ a) Resets the Arduino
- ☐ b) Turns the LED off
- ☒ c) Prepares the sensor for a new ping
- ☐ d) Measures temperature

5. Which component is not part of this project?

- ☒ a) Servo motor
- ☐ b) Ultrasonic sensor
- ☒ c) LED
- ☐ d) Resistor

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6. How would you make the LED stay on longer after detection?

- ☒ a) Replace the LED with a buzzer
- ☐ b) Remove the delay()
- ☐ c) Use analogWrite()
- ☐ d) Increase the delay() after digitalWrite(LED_PIN, HIGH)

7. What happens if an object is 30 cm away?

- ☒ a) LED stays off
- ☐ b) LED blinks
- ☐ c) Servo moves
- ☐ d) Buzzer sounds

8. Which Arduino function measures the ultrasonic pulse duration?

- ☒ a) delayMicroseconds()
- ☐ b) pulseIn()
- ☐ c) analogRead()
- ☐ d) tone()

9. What is the purpose of the 220-ohm resistor?

- ☒ a) Protect the LED from excess current
- ☐ b) Calibrate the ultrasonic sensor
- ☐ c) Power the buzzer
- ☐ d) Measure distance

10. How could you add a buzzer to alert when the light turns on?

- ☒ a) Connect a buzzer to another pin and activate it with the LED
- ☐ b) Replace the LED with a buzzer
- ☐ c) Use tone() instead of digitalWrite()
- ☐ d) Both a and c

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Project 4: LDR-Based 4 LED Bar Graph

1. What does the LDR measure in this project?

- ☐ a) Temperature
- ☒ b) Light intensity
- ☐ c) Distance
- ☐ d) Sound level

2. How are the LEDs arranged to show light levels?

- ☐ a) In parallel with independent resistors
- ☒ b) In series with one resistor
- ☐ c) Without any resistors
- ☐ d) Using PWM only

3. What does `lightLevel > ledThresholds[i]` determine?

- ☒ a) Which LEDs turn on based on light intensity
- ☐ b) The angle of a servo
- ☐ c) The frequency of a buzzer
- ☐ d) The text on an LCD

4. Which Arduino function reads the LDR value?

- ☐ a) `tone()`
- ☐ b) `digitalRead()`
- ☐ c) `pulseIn()`
- ☒ d) `analogRead()`

5. How would you adjust the project for a darker room?

- ☒ a) Decrease the threshold values in `ledThresholds[]`
- ☐ b) Increase the resistor value
- ☐ c) Remove the LDR
- ☐ d) Use `digitalWrite()` instead

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6. What is the purpose of the 220-ohm resistors?

- ☐ a) Power the Arduino
- ☐ b) Calibrate the LDR
- ☒ c) Limit current to the LEDs
- ☐ d) Measure voltage

7. Which component is not used?

- ☒ a) Buzzer
- ☐ b) LDR
- ☒ c) LEDs
- ☐ d) Resistors

8. What happens if all 4 LEDs are lit?

- ☐ a) The buzzer sounds
- ☐ b) The LDR is broken
- ☒ c) The Arduino is overheating
- ☒ d) The environment is very bright

9. How is the LDR connected to the Arduino?

- ☒ a) As a voltage divider with a 10k-ohm resistor
- ☒ b) Directly to 5V
- ☐ c) To a digital pin
- ☐ d) To a servo

10. What does map() do in this project?

- ☐ a) Not used (thresholds are fixed)
- ☒ b) Converts LDR values to LED counts
- ☒ c) Controls servo angle
- ☐ d) Generates sound

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Project 5: Morse Code Decoder

1. What input method is used for Morse code?

- ☐ a) LDR
- ☐ b) Ultrasonic sensor
- ☒ c) Two buttons (dot/dash)
- ☐ d) Potentiometer

2. How is the Morse code "A" (.-) represented in the code?

- ☐ a) if (morseCode == ".-") return 'A'
- ☐ b) tone(buzzerPin, 262)
- ☒ c) digitalWrite(LED_PIN, HIGH)
- ☐ d) analogRead(A0)

3. What does the enterButton do?

- ☐ a) Adjusts brightness
- ☐ b) Resets the Arduino
- ☒ c) Controls servo angle
- ☐ d) Submits the Morse sequence for decoding

4. Which library is used for the LCD?

- ☒ a) LiquidCrystal_I2C.h
- ☐ b) Servo.h
- ☒ c) NewPing.h
- ☐ d) TM1637.h

5. How is the decoded character displayed?

- ☐ a) Via serial monitor
- ☒ b) On the LCD screen
- ☐ c) Through LED blinks
- ☒ d) Using a buzzer tone

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6. What is the purpose of debounce()?

- ☒ a) Prevents multiple button press registrations
- ☒ b) Measures distance
- ☐ c) Controls servo speed
- ☐ d) Adjusts LDR sensitivity

7. Which component provides user feedback?

- ☐ a) LCD
- ☐ b) Servo
- ☒ c) Ultrasonic sensor
- ☐ d) Temperature sensor

8. How would you add sound feedback for dots/dashes?

- ☐ a) Ignore the buttons
- ☒ b) Replace the LCD with a buzzer
- ☐ c) Use tone() with short/long beeps
- ☐ d) Use analogWrite()

9. What is stored in morseCode?

- ☒ a) The sequence of dots/dashes (e.g., ".-")
- ☐ b) The decoded letter
- ☐ c) The LED brightness
- ☐ d) The servo angle

10. How is the letter "B" (---..) mapped?

- ☒ a) if (morseCode == "---..") return 'B'
- ☐ b) digitalWrite(2, HIGH)
- ☐ c) analogRead(A0)
- ☐ d) delay(10)

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Project 6: Simple Digital Clock

1. Which module provides accurate timekeeping?

- ☒ a) DS3231 RTC
- ☐ b) Ultrasonic sensor
- ☐ c) LDR
- ☐ d) Potentiometer

2. How is the time displayed?

- ☐ a) On an LCD screen
- ☐ b) Via 4-digit display
- ☒ c) Through LED blinks
- ☐ d) Using a buzzer

3. What does `rtc.now()` return?

- ☒ a) Current date/time
- ☐ b) Light level
- ☐ c) Distance in cm
- ☐ d) Temperature

4. Which library is not needed?

- ☒ a) `RTClib.h`
- ☐ b) `Wire.h`
- ☒ c) `Servo.h`
- ☐ d) `LiquidCrystal_I2C.h`

5. How often does the clock update?

- ☒ a) Every second (`delay(1000)`)
- ☐ b) Every minute
- ☒ c) Only when reset
- ☐ d) Continuously without delay

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6. What happens if the RTC loses power?

- ☐ a) The LCD stops working
- ☒ b) It may need time resetting (rtc.adjust())
- ☒ c) The buzzer sounds
- ☐ d) The LEDs blink

7. Which pins connect the RTC to the Arduino?

- ☒ a) SDA (A4) and SCL (A5)
- ☒ b) D2 and D3
- ☒ c) 5V and GND
- ☐ d) Trig and Echo

8. How would you add an alarm feature?

- ☐ a) Use pulseIn()
- ☐ b) Replace the RTC with an LDR
- ☒ c) Compare rtc.now() to a set time and trigger a buzzer
- ☐ d) Ignore the LCD

9. What is the purpose of lcd.print("Time: ")?

- ☐ a) Labels the time display
- ☒ b) Resets the clock
- ☐ c) Measures light
- ☐ d) Controls a servo

10. How is the date formatted on the LCD?

- ☐ a) second/minute/hour
 - ☒ b) month/day/year
 - ☐ c) year/month/day
 - ☐ d) day/month/year
-

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Project 7: Automatic Gate Opener

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1. What triggers the gate to open?

- ☐ a) Button press
- ☐ b) High temperature
- ☐ c) Bright light
- ☒ d) Object detected within 20cm by ultrasonic sensor

2. How is the gate simulated?

- ☒ a) Servo motor (0° = closed, 90° = open)
- ☐ b) LED brightness
- ☐ c) Buzzer tone
- ☐ d) LCD message

3. Which library controls the servo?

- ☐ a) NewPing.h
- ☒ b) Servo.h
- ☒ c) LiquidCrystal_I2C.h
- ☐ d) RTCLib.h

4. What does sonar.ping_cm() measure?

- ☒ a) Distance to object
- ☐ b) Light intensity
- ☐ c) Temperature
- ☐ d) Sound frequency

5. How is the gate status shown?

- ☐ a) Buzzer beeps
- ☒ b) LED blinks
- ☐ c) LCD displays "Gate Open/Closed"
- ☐ d) Serial monitor

6. What happens if no object is detected?

- ☒ a) Servo returns to 0° (closed)
- ☐ b) LCD turns off
- ☐ c) Buzzer sounds continuously
- ☐ d) LED stays on

7. Which component is not used?

- ☒ a) LDR
- ☐ b) Ultrasonic sensor
- ☐ c) Servo
- ☐ d) LCD

8. How would you increase the detection range?

- ☐ a) Remove the servo
- ☒ b) Adjust if (distance < 30)
- ☐ c) Use analogRead()
- ☐ d) Replace the sensor with an LDR

9. What is the role of NewPing.h?

- ☒ a) Simplifies ultrasonic sensor code
- ☐ b) Controls LCD brightness
- ☐ c) Measures temperature
- ☐ d) Generates sound

10. How could you add a manual override button?

- ☒ a) Add a button to toggle the servo with digitalRead()
 - ☐ b) Replace the ultrasonic sensor
 - ☐ c) Ignore the LCD
 - ☐ d) Use tone()
-

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Project 8: Motion Detector Alarm System

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1. Which sensor detects motion?

- ☒ a) PIR sensor
- ☐ b) Ultrasonic sensor
- ☐ c) LDR
- ☐ d) Potentiometer

2. How is the alarm signaled?

- ☒ a) Buzzer sounds and LED turns on
- ☐ b) LCD shows temperature
- ☒ c) Servo moves
- ☐ d) LEDs dim

3. What does digitalRead(pirPin) == HIGH indicate?

- ☐ a) Low light
- ☐ b) No motion
- ☒ c) Motion detected
- ☐ d) High temperature

4. Which component provides visual feedback?

- ☒ a) LED
- ☐ b) Servo
- ☐ c) Potentiometer
- ☐ d) RTC

5. How would you adjust the sensor's sensitivity?

- ☐ a) Use analogWrite()
- ☐ b) Replace the PIR with an LDR
- ☒ c) Change delay(1000)
- ☐ d) Use the potentiometer on the PIR module

6. What is the purpose of lcd.clear()?

- ☒ a) Reset the display before updating
- ☐ b) Turn off the buzzer
- ☒ c) Calibrate the sensor
- ☐ d) Measure distance

7. Which library is used for the LCD?

- ☐ a) Servo.h
- ☒ b) LiquidCrystal_I2C.h
- ☐ c) NewPing.h
- ☐ d) RTCLib.h

8. How could you make the alarm delay before triggering?

- ☒ a) Add delay(2000) after motion detection
- ☐ b) Remove the PIR sensor
- ☒ c) Use tone()
- ☐ d) Ignore the LED

9. What happens when motion stops?

- ☐ a) LED and buzzer turn off
- ☒ b) LCD shows "No Motion"
- ☒ c) Both a and b
- ☐ d) Servo moves to 90°

10. How would you add a reset button?

- ☒ a) Connect a button to reset the alarm state
 - ☐ b) Replace the PIR sensor
 - ☐ c) Use analogRead()
 - ☐ d) Remove the LCD
-

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AI & You Chapter 1: How AI Solves Everyday Problems

1. In Ramesh's story, how did the AI farming app help him?

- ☐ a) By watering crops automatically
☒ b) By scanning plants to detect pests
☐ c) By driving his tractor
☐ d) By predicting the price of crops

2. Which of these is a real-world application of AI in medicine?

- ☐ a) Analyzing X-rays to diagnose diseases
☒ b) Washing hospital floors
☐ c) Turning lights on/off
☐ d) Counting pills manually

3. What is the primary feature of AI mentioned in the chapter?

- ☐ a) It follows fixed instructions without learning
☐ b) It analyzes data, finds patterns, and adapts
☐ c) It only works with robots
☐ d) It replaces all human jobs

4. Where might you find AI in your neighborhood?

- ☐ a) Smart streetlights that adjust brightness
☒ b) A manual bicycle
☐ c) A traditional wall clock
☐ d) A pencil

5. What activity does the chapter suggest to "Spot AI in Action"?

- ☐ a) Drawing a picture of a computer
☐ b) Building a cardboard robot
☒ c) Observing smart devices like cameras or voice assistants
☐ d) Memorizing math formulas

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6. Which of these is NOT an example of AI?

- ☐ a) A washing machine with a start button
- ☐ b) A security camera that detects movement
- ☒ c) A smartphone app that recommends videos
- ☐ d) A self-driving car

7. How does AI improve transportation systems?

- ☐ a) By controlling traffic lights based on real-time traffic
- ☐ b) By painting road signs
- ☒ c) By replacing all buses with bicycles
- ☐ d) By selling tickets manually

8. What is the purpose of the "AI or Not AI" challenge?

- ☐ a) To teach Morse code
- ☐ b) To build a circuit with LEDs
- ☒ c) To measure temperature with a sensor
- ☐ d) To distinguish between adaptive AI and static machines

9. What role does AI play in environmental protection?

- ☐ a) Monitoring pollution levels and predicting wildfires
- ☒ b) Planting trees manually
- ☐ c) Cleaning ocean plastic by hand
- ☐ d) Turning off lights in empty rooms

10. What is the key takeaway about AI's limitations?

- ☐ a) It doesn't require data to learn
- ☐ b) It can solve all problems perfectly
- ☒ c) It still needs human oversight and guidance
- ☐ d) It works without electricity

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AI & You – Chapter 2: How AI Learns and Gets Smarter

1. How did the AI app respond when the girl did well?

- ☐ a) By giving harder questions when she excelled
- ☒ b) By shutting down after 10 minutes
- ☐ c) By ignoring her mistakes
- ☐ d) By repeating the same questions daily

2. What is the core idea behind how AI learns?

- ☒ a) It improves by analyzing feedback and practicing
- ☐ b) It only works with numbers, not words
- ☐ c) It requires manual reprogramming daily
- ☐ d) It never changes its responses

3. Which example shows AI "getting smarter" over time?

- ☐ a) A toaster browning bread the same way every day
- ☐ b) A voice assistant understanding accents better with use
- ☒ c) A fan spinning at fixed speeds
- ☐ d) A light bulb burning out after a year

4. What is the goal of the "Help AI Learn New Skills" activity?

- ☒ a) To build a paper airplane
- ☐ b) To observe how apps adjust difficulty based on performance
- ☒ c) To memorize a list of animals
- ☐ d) To measure the brightness of an LED

5. In the "Fun AI Experiment," what does testing Google Lens demonstrate?

- ☐ a) The speed of a servo motor
- ☐ b) How to solder components
- ☒ c) AI's ability to recognize objects in images
- ☐ d) The resistance of an LDR

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6. Why might an AI chatbot sometimes give incorrect answers?

- ☒ a) It only responds to voice commands
- ☐ b) It refuses to learn new things
- ☐ c) It relies on patterns in data, which can be incomplete
- ☐ d) It doesn't use electricity

7. What is a real-world example of AI in games?

- ☒ a) A dice-rolling app
- ☐ b) A static puzzle with fixed solutions
- ☐ c) An opponent that becomes harder as you improve
- ☐ d) A manual scoreboard

8. How does AI assist doctors, as mentioned in the chapter?

- ☐ a) By replacing all nurses
- ☒ b) By cleaning surgical tools
- ☐ c) By analyzing medical scans for abnormalities
- ☐ d) By scheduling appointments on paper

9. What does the "Create Your Own AI Tool" task encourage?

- ☒ a) Drawing a map of your school
- ☐ b) Brainstorming ideas for AI applications in daily life
- ☐ c) Memorizing the periodic table
- ☐ d) Building a circuit with resistors

10. What is the summary message about AI's learning process?

- ☐ a) It only works in laboratories
- ☐ b) It doesn't need human interaction
- ☒ c) It mimics human learning but requires data and feedback
- ☐ d) It is perfect from the first use